



Department of Computer Science and Engineering
Final Examination Fall 2022
CSE 321: Operating Systems

Duration: 2 Hours

Total Marks: 40

Answer the following questions.
Figures in the right margin indicate marks.

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1. a) A system has processes to execute of which **30%** is serial. If the number of
CO4 cores is increased from **1** to **3**, what will be the **increase** in performance? [2]
b) **Distinguish** between **many-to-one** and **one-to-one** multithreading models. [2]
2. a) Suppose a medical center is providing Covid vaccination. In that center [2]
CO5 maximum of 6 people can take vaccines at a time in separate booths. But approximately 50 people went there to take vaccines on a particular day. Therefore, the authorities have decided that they will provide vaccines to 6 people at a time and keep others waiting in a queue. If any of the vaccine booths get free a person from the queue will be taken to that booth according to the first come first serve manner for vaccination. **Logically explain** which synchronization method has been used here
- b) For Peterson's problem below conditions will be applied.
- There are two processes: P1 and P2.
 - Each Statement takes 4ms to execute.
 - Context Switch will occur after 12ms.
 - Both the Critical & Remainder section contains 3 statements.
 - For P1: i=0 and j=1
 - For P2: i=1 and j=0
 - turn=0
 - flag[0] = FALSE, flag[1] = FALSE

The structure of process P_i in Peterson's solution:

```
do{
    flag[i] = true;
    turn = j;
    while(flag[j] == true && turn == 1){
        //busy wait
    }
    //critical section
    flag[i] = false;
    //remainder section
}while(true);
```

Complete the table given below for processes P1 and P2 using **Peterson's solution**.

[4]

Process 1: i=0, j=1	Process 2: i=1, j=0
↓	

3. a) Consider the following snapshot of a system:
C05

		Allocation				MAX			
		A	B	C	D	A	B	C	D
P ₀		5	1	1	4	10	5	10	11
P ₁		2	8	6	4	9	12	9	10
P ₂		2	2	6	2	5	9	10	3
P ₃		4	6	8	2	4	6	8	3

Available			
4	4	6	4

- i. Is the system in a safe state? **Apply Banker's safety algorithm** to find out the safe sequence. You need to calculate the need matrix. [4]
- ii. P₀ requests for **(3 3 6 1)**, **check** the validity of the request. If the request is valid, does the system enter a **deadlock**? [2+3]
- b) Suppose, in an office, we have a set of resource types, R = {R1, R2, R3} and a set of processes, P = {P1, P2, P3, P4, P5}. **R1, R2, and R3** have **3, 2, and 1** instance respectively.

- P1 is holding 1 instance of R3
- P1 requests 1 instance of R1
- P3 requests for 1 instance of R3
- P3 is holding 2 instances of R2
- P2 requests for 1 instance of R2
- P2 is holding 1 instance of R1
- P4 is holding 1 instance of R1
- P4 requests 1 instance of R2
- P5 is holding 1 instance of R1

[3]

Construct a resource allocation graph for the above scenario. **Mention** the number of cycles found and **identify** whether there is a deadlock or not.

4. a) At a particular time, the snapshot of the Main memory is given below for dynamic partitioning where gray portions of the memory represent occupied spaces. **Apply worst fit** and **first fit** algorithms to place processes with the space requirement of P1=300k, P2=200k, P3=149k, P4=146k, P5=100k, P6=50k, P7=22k and P8=29k (in order). **Explain** which algorithm makes the most effective use of memory. [2+2+1]

400K	300K	160K	50K	200K	260K
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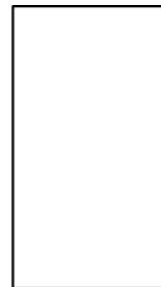
- b) Assume that in a paged memory management system the page size for processes is 4 bytes and the Physical Memory size is 36 bytes. **Show** the users' view of memory which is mapped into physical memory. [2]

P0	Free
P1	Finite
P2	Fruit
P3	From
P4	Flower
P5	Freedom

Logical space of a process

0	2
1	7
2	4
3	5
4	3
5	1

PMT



Main Memory

- c) If the page size is **6 KB**, **how many** frames will be needed in Main memory for a process size of **102,506 Bytes**? Is there any **internal fragmentation**? - If yes, **calculate** the value. [1 KB = 1024 Bytes] [2]
- d) **Discuss** the purpose of **MMU**. [3]

5. Consider a computer with a main memory that has 3 frames and page reference string of 0-7 pages: **[3 0 6 4 2 6 7 2 0 1 7]**. The page reference string represents the order in which the pages are accessed by a program. **Apply FIFO & LRU** algorithm to simulate the page replacement that occurs when the main memory can hold at most 3 pages at a time. **Record** the number of **page faults** and compare the result. **Mention** which algorithm performs better in this scenario. [4+1+1]